

pct_oi_within 5pct & 10pct - v4

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Research Question

please take a look at pct_oi_within_5pct_co and it's 10_pct variant (you can ignore the "-pc" variations) and explore all tickers individually to see if specific tickers show promise for predictive traits over the next 1 & 5 days. Provide a wide variety of visuals to illustrate your findings, including a graphic of the score for each ticker.

Table: daily_features

Tickers: AAL, AAPL, ADBE, AMAT, AMC, AMD, AMZN, AVGO, BA, BABA, BAC, C, COIN, COST, CRWD, CVNA, CVS, F, GLD, GOOGL, HOOD, IBIT, INTC, IWM, JPM, KO, KRE, LLY, MARA, META, MRVL, MSFT, MSTR, MU, NFLX, NIO, NVDA, ORCL, PFE, PLTR, QQQ, RIOT, SBUX, SLV, SMCI, SOFI, SPOT, SPY, TSLA, TSM, UBER, UNH, UPS, USO, V, VXX, WMT, XBI, XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XOM, XOP

Date range: 2020-01-01 to 2026-04-30

Model: claude-sonnet-4-6

Workflow Plan

Task Type: single-factor-scan

Depth: broad

Reasoning: The user wants to evaluate two specific OI concentration metrics (pct_oi_within_5pct_co and pct_oi_within_10pct_co) as predictive signals for forward returns at 1-day and 5-day horizons, analyzed individually per ticker — this is a classic single-factor scan across the ticker universe.

Hypotheses:

- Higher pct_oi_within_5pct_co (more OI concentrated near the spot price on the combined basis) predicts positive 1-day forward returns for specific tickers, suggesting a gravitational/pinning effect.
- Higher pct_oi_within_10pct_co predicts positive 5-day forward returns for specific tickers, implying a broader near-money OI concentration acts as a medium-term magnet.
- The predictive strength of pct_oi_within_5pct_co and pct_oi_within_10pct_co varies significantly across tickers, with some showing statistically meaningful IC (information coefficient) and others showing no relationship.
- Extreme quantile readings (top/bottom decile) of pct_oi_within_5pct_co generate asymmetric return distributions, revealing non-linear predictive dynamics for certain tickers.
- The 5pct variant captures sharper pinning effects while the 10pct variant provides a smoother, more persistent signal over the 5-day horizon.

Feature columns (X): pct_oi_within_5pct_co, pct_oi_within_10pct_co, put_call_oi_ratio, pct_oi_in_front_expiry, oi_above_below_ratio_co, zscore_put_call_oi_ratio_3m, d1_total_oi_pct_change, d5_total_oi_pct_change, rv_5d, rv_20d, weighted_avg_dte, zscore_oi_above_below_ratio_3m_co, top5_strikes_pct_total_oi, oi_within_5pct_co

Outcome columns (Y): ret_1d_fwd_oc, ret_5d_fwd_oc

Key Questions:

- Which tickers show the highest absolute Spearman IC between pct_oi_within_5pct_co and ret_1d_fwd_oc?
- Which tickers show the highest absolute Spearman IC between pct_oi_within_10pct_co and ret_5d_fwd_oc?

- Is the IC statistically significant ($|t\text{-stat}| > 2$) for any ticker at either horizon?
- Do the top and bottom deciles of `pct_oi_within_5pct_co` produce meaningfully different forward return distributions for any ticker?
- Is the 10pct variant a stronger predictor than the 5pct variant over the 5-day horizon across most tickers?
- Are there tickers where both features (5pct and 10pct) consistently score high, suggesting robust near-money OI concentration signals?
- What composite predictive score (e.g., combining IC magnitude, t-stat, and hit rate) does each ticker receive, and can this be visualized in a ranked graphic?
- Does the signal direction (positive vs negative IC) differ systematically across tickers — e.g., mean-reversion vs momentum character?
- How stable is the rolling IC over time for the highest-scoring tickers?
- Is there a relationship between realized volatility (`rv_5d`) and the strength of the `pct_oi_within_5pct_co` signal?

Analysis Focus: Per-ticker information coefficient (IC) and rank IC between `pct_oi_within_5pct_co` / `pct_oi_within_10pct_co` and forward returns (1d, 5d); quantile return spreads (top vs bottom decile of each feature); hit rate, t-stat, and composite predictive score per ticker; cross-ticker ranking of signal quality; non-linear (quantile-binned) return profiles; rolling IC stability over time per ticker.

Research Report

Best Composite Score by Ticker

	Composite Score
AAPL	82.500
XLB	77.700
V	75.300
C	75.100
VXX	74.400
XLV	73.800
TSLA	73.700
XLP	73.500
MSFT	73.400
AVGO	73.300
KO	73.300
SPY	72.400
WMT	72.000
IWM	71.400
MARA	70.500
MSTR	70.500

AMD	70.000
TSM	69.800
KRE	69.500
QQQ	68.300
ORCL	68.000
SBUX	67.600
XLF	67.600
CVNA	67.200
BA	67.100
USO	66.900
GOOGL	66.700
ADBE	66.500
XLI	66.500
COIN	66.400
...	...

Decile Profile: TSLA | pct_oi_within_5pct_co → ret_5d_fwd_oc (score=74)

	Avg Return %
D1	4.733
D2	1.502
D3	-0.486
D4	1.808
D5	1.706
D6	0.387
D7	0.493
D8	0.410
D9	0.079
D10	-0.187

Decile Profile: ORCL | oi_within_5pct_co → ret_5d_fwd_oc (score=68)

	Avg Return %
D1	1.469
D2	0.474
D3	0.460
D4	0.360
D5	0.800

D6	0.749
D7	0.482
D8	0.080
D9	0.553
D10	-1.349

Decile Profile: ADBE | oi_within_5pct_co → ret_5d_fwd_oc (score=66)

	Avg Return %
D1	1.394
D2	0.974
D3	0.274
D4	-0.820
D5	-0.269
D6	0.881
D7	0.436
D8	-0.985
D9	-1.237
D10	-0.361

Decile Profile: CRWD | pct_oi_within_5pct_co → ret_5d_fwd_oc (score=66)

	Avg Return %
D1	3.511
D2	1.780
D3	1.212
D4	0.642
D5	1.292
D6	0.097
D7	-0.341
D8	0.577
D9	-0.655
D10	0.900

Decile Profile: MRVL | pct_oi_within_5pct_co → ret_5d_fwd_oc (score=66)

	Avg Return %
D1	3.016

D2	1.486
D3	-0.039
D4	1.063
D5	0.415
D6	1.522
D7	0.766
D8	-0.815
D9	-0.647
D10	0.250

Decile Profile: TSLA | oi_within_5pct_co → ret_5d_fwd_oc (score=65)

	Avg Return %
D1	5.351
D2	0.907
D3	-0.880
D4	2.080
D5	1.369
D6	0.008
D7	0.543
D8	0.615
D9	1.615
D10	-1.172

Decile Profile: AAPL | pct_oi_within_5pct_co → ret_5d_fwd_oc (score=65)

	Avg Return %
D1	1.808
D2	1.156
D3	0.181
D4	0.646
D5	-0.487
D6	0.088
D7	0.759
D8	0.126
D9	0.399
D10	0.335

Decile Profile: AAPL | oi_within_5pct_co → ret_5d_fwd_oc (score=64)

	Avg Return %
D1	1.607
D2	1.234
D3	0.509
D4	0.413
D5	0.209
D6	0.495
D7	-0.132
D8	0.034
D9	0.270
D10	0.372

Equity Curve: AAPL | oi_above_below_ratio_co → ret_5d_fwd_oc

	Top (top)	Bottom (bottom)
2020-01	2.350	-15.020
2020-01	8.060	-15.890
2020-01	10.330	-12.380
2020-01	7.730	-9.900
2020-01	7.560	-13.320
2020-02	12.720	-14.420
2020-02	16.150	-13.830
2020-02	10.710	-19.710
2020-02	9.990	-16.550
2020-03	-2.350	-16.420
2020-05	1.290	-16.770
2020-05	1.160	-14.130
2020-05	2.990	-17.630
2020-05	2.480	-18.160
2020-06	6.810	-15.660
2020-06	9.400	-13.450
2020-06	14.340	-9.510
2020-06	14.990	-15.020
2020-06	20.820	-13.690
2020-07	24.520	-12.660
2020-07	23.550	-15.840

2020-07	19.610	-21.460
2020-07	33.000	-20.470
2020-08	35.700	-17.830
2020-08	37.740	-20.840
2020-08	44.900	-24.320
2020-08	41.880	-24.370
2020-08	36.690	-22.360
2022-05	35.320	-22.930
2022-05	35.780	-24.850
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Equity Curve: XLB | pct_oi_in_front_expiry → ret_5d_fwd_oc

	Top (top)	Bottom (bottom)
2020-01	2.530	-3.610
2020-06	7.370	-3.020
2020-12	8.500	3.170
2021-01	8.440	6.020
2021-06	2.460	9.970
2022-06	-3.400	13.370
2022-11	-4.620	11.530
2023-01	-5.670	15.600
2023-09	-6.420	13.760
2024-01	-6.990	16.890
2024-05	-7.700	17.340
2024-05	-8.130	13.750
2024-06	-10.010	23.180
2024-06	-11.440	25.580
2024-07	-10.340	27.790
2024-07	-11.820	28.470
2024-08	-9.450	30.230
2024-08	-8.220	24.660
2024-09	-10.540	27.190
2024-09	-7.450	27.400
2024-09	-9.100	27.990
2024-10	-7.880	27.500
2024-10	-5.610	25.810

2024-10	-7.750	22.400
2024-11	-4.860	22.800
2024-11	-4.600	23.080
2024-12	-6.040	23.520
2024-12	-6.800	26.870
2025-01	-6.920	25.640
2025-01	-6.960	26.410
...	...	...

Equity Curve: V | oi_above_below_ratio_co → ret_5d_fwd_oc

	Top (top)	Bottom (bottom)
2020-02	6.680	3.520
2020-03	9.380	8.890
2020-03	6.530	7.690
2020-03	20.840	6.930
2020-03	15.140	8.710
2020-04	21.160	3.380
2020-10	23.530	5.870
2021-01	26.050	3.910
2021-02	33.030	4.080
2021-09	39.770	2.510
2021-10	34.690	0.330
2021-11	38.130	-1.720
2021-11	38.890	-9.230
2021-11	32.920	-8.020
2021-11	31.060	-7.150
2021-11	28.730	-7.360
2021-12	36.090	-6.890
2022-01	48.860	-12.080
2022-02	51.380	-13.530
2022-03	44.040	-13.850
2022-03	42.470	-15.270
2022-03	52.900	-14.110
2022-04	56.180	-13.260
2022-05	48.650	-13.330
2022-05	48.240	-12.140

2022-05	49.110	-12.240
2022-06	42.860	-12.580
2022-06	41.760	-11.340
2022-06	45.770	-10.710
2022-06	47.690	-7.790
...	...	...

Yearly Consistency: AAPL | oi_above_below_ratio_co → ret_5d_fwd_oc

	Top Decile	Bottom Decile
2020	2.370	-0.984
2021	0.512	-0.266
2022	0.925	-0.851
2023	1.832	-0.552
2024	0.858	0.277
2025	2.744	-0.273
2026	1.022	-2.847

D10 vs D1 Win Rate by Ticker (best signal per ticker)

	D10 Win Rate %	D1 Win Rate %
AAL	56.300	42.100
AAPL	72.500	39.600
ADBE	46.200	63.500
AMAT	51.300	69.800
AMC	41.800	37.100
AMD	61.400	49.100
AMZN	56.700	44.900
AVGO	70.500	28.900
BA	44.900	50.900
BABA	50.000	46.500
BAC	34.800	52.200
C	35.400	58.500
COIN	42.000	57.500
COST	71.500	54.100
CRWD	56.300	61.000
CVNA	55.700	35.900
CVS	61.400	40.900

F	51.900	57.200
GLD	67.700	50.300
GOOGL	55.700	65.400
HOOD	44.100	52.900
IBIT	60.000	47.200
INTC	46.800	54.100
IWM	55.100	61.600
JPM	52.500	54.100
KO	61.400	39.600
KRE	37.300	62.900
LLY	58.700	39.300
MARA	47.100	45.500
META	60.400	51.800
...	...	...

D10 minus D1 Return Spread by Ticker (%)

	Spread %
MARA	9.993
AVGO	8.554
CVNA	8.358
USO	6.000
RIOT	5.881
NIO	5.292
XOP	3.832
PLTR	3.780
AMC	3.302
AMD	3.190
AAPL	3.140
SMCI	3.091
SLV	1.990
LLY	1.977
AMZN	1.897
V	1.829
IBIT	1.778
AAL	1.711
UNH	1.670

XLV	1.609
XLF	1.491
XLE	1.174
SBUX	1.117
COST	0.799
GLD	0.725
MSFT	0.719
SPOT	0.641
XLP	0.621
SPY	0.496
CVS	0.467
...	...

Caveats & Challenges

Skeptic Review

Sample Size & Precision

Bucket sizes of ~153–158 observations appear adequate at face value, but these are *daily* observations from a 7-year window, meaning heavy autocorrelation inflates the effective N dramatically. With 5-day forward returns, adjacent observations share roughly 80% of their return window. True independent observations are closer to N/5 (~300 total), yielding ~30 per decile bucket — well below any threshold for reliable inference. The precision implied by scores like 82 or 75 is not credible under this correction.

100% Consistency on 7 Years

Seven yearly periods is a thin basis for claiming "100% consistency." With seven binary outcomes, random chance produces 100% directional consistency roughly 1-in-128 times — not negligible when screening dozens of ticker-feature pairs. Multiple comparisons across ~8 reported signals make at least one spurious 100%-consistent signal highly probable.

Tail Concentration

Mid-decile ranges of 0.0031–0.0121 suggest the monotonic story may hold in aggregate, but without interior decile breakdowns, concentration in the extreme deciles masking a flat middle cannot be ruled out. XLB's `rv_20d` mid-decile range of 0.0031 is economically negligible after costs.

Drawdown & Tradeability

AAPL bottom bucket: maxDD of -38.0% on a 46% win rate is not a tradeable signal — it requires tolerating near-drawdown-to-ruin conditions. XLB top bucket: -6.3% cumulative return with -20.3% maxDD at a 42% win rate has a negative Calmar ratio and should be disqualified outright.

OI-as-Dealer-Gamma Conflation

`oi_above_below_ratio_co` and `pct_oi_in_front_expiry` are OI-derived constructs. They reflect aggregate open interest structure, not verified dealer inventory. Treating them as dealer gamma signals violates basic data limitations — these should be labeled OI-implied proxies only.

Regime Risk

No regime decomposition is presented. Seven years (roughly 2017–2024) includes a historically anomalous zero-rate period and a volatility shock regime. Signals showing 85.7% consistency (6/7 years) may be masking a single dominant regime driving most of the edge.

Scan Results

Score	Ticker	Feature	Outcome	Pattern	Consistency	N
82	AAPL	oi_above_below_ratio_co	ret_5d_fwd_oc	linear_weak_pos	100.0%	1536
78	XLB	pct_oi_in_front_expiry	ret_5d_fwd_oc	linear_weak_neg	100.0%	1585
75	V	oi_above_below_ratio_co	ret_5d_fwd_oc	linear_weak_pos	100.0%	1585
75	XLB	rv_20d	ret_5d_fwd_oc	linear_weak_pos	85.7%	1585
75	C	put_call_oi_ratio	ret_5d_fwd_oc	inverted_u	85.7%	1585
74	VXX	put_call_oi_ratio	ret_5d_fwd_oc	threshold	85.7%	1582
74	XLV	oi_above_below_ratio_co	ret_5d_fwd_oc	threshold	100.0%	1585
74	TSLA	pct_oi_within_5pct_co	ret_5d_fwd_oc	threshold	100.0%	1585
74	XLP	oi_above_below_ratio_co	ret_5d_fwd_oc	linear_weak_pos	85.7%	1585
73	MSFT	zscore_put_call_oi_ratio_	ret_5d_fwd_oc	u_shape	85.7%	1585
73	AVGO	oi_above_below_ratio_co	ret_5d_fwd_oc	threshold	100.0%	445
73	KO	pct_oi_within_10pct_co	ret_1d_fwd_oc	monotonic_posit	100.0%	1589
72	SPY	weighted_avg_dte	ret_5d_fwd_oc	u_shape	85.7%	1585
72	V	zscore_oi_above_below_rat	ret_5d_fwd_oc	linear_weak_pos	100.0%	1585
72	WMT	zscore_oi_above_below_rat	ret_5d_fwd_oc	inverted_u	100.0%	482
71	IWM	zscore_put_call_oi_ratio_	ret_5d_fwd_oc	linear_weak_neg	85.7%	1585
71	XLP	put_call_oi_ratio	ret_5d_fwd_oc	u_shape	71.4%	1585
70	MARA	top5_strikes_pct_total_oi	ret_5d_fwd_oc	linear_weak_pos	85.7%	1536
70	MSTR	rv_20d	ret_5d_fwd_oc	threshold	71.4%	1585
70	AMD	weighted_avg_dte	ret_5d_fwd_oc	threshold	100.0%	1585
70	TSM	pct_oi_in_front_expiry	ret_5d_fwd_oc	inverted_u	100.0%	1585
70	KRE	put_call_oi_ratio	ret_5d_fwd_oc	threshold	85.7%	1585
68	QQQ	put_call_oi_ratio	ret_5d_fwd_oc	threshold	100.0%	1585
68	KO	oi_above_below_ratio_co	ret_5d_fwd_oc	isolated_region	85.7%	1585
68	ORCL	oi_within_5pct_co	ret_5d_fwd_oc	threshold	85.7%	1585
68	XLP	zscore_put_call_oi_ratio_	ret_5d_fwd_oc	u_shape	71.4%	1585
68	SBUX	oi_above_below_ratio_co	ret_5d_fwd_oc	threshold	100.0%	1585
68	XLF	oi_above_below_ratio_co	ret_5d_fwd_oc	threshold	100.0%	1585
68	QQQ	rv_5d	ret_5d_fwd_oc	threshold	100.0%	1585
67	CVNA	weighted_avg_dte	ret_5d_fwd_oc	threshold	71.4%	1584